



Department of Physics
National Institute of Technology Kurukshetra

Course: B.Tech(CoE, IT, AI&ML, IIOT, M&C, M&V, AI&DS)

Exam: Sessional-I

Date: 19-09-2024

Subject: Engineering Physics

Course Code: PHIC 101

Duration: 50 min

<i>Note: All questions are compulsory.</i>		<i>Max. Marks: 20</i>	<i>Roll No:</i>
1.(a) Explain, with a clear and relevant diagram, the processes of stimulated absorption, spontaneous emission, and stimulated emission of light in the atomic system.			5
2.(a) Describe the structure of an optical fiber and explain its working principle, supported by a diagram. (b) The refractive indices of the core and cladding of an optical fiber are 1.55 and 1.5, respectively. If light enters the fiber at an angle of 10° with respect to the fiber axis, evaluate whether the light will be guided and confined within the core. (<i>Refractive index of air=1</i>)			3+2
3.(a) What is length contraction, and derive its expression? If a cube with side length L is moving with relativistic velocity along the y-direction, in which direction does length contraction occur? (b) If a spacecraft is moving at $0.99c$ toward a planet, how would the planet's diameter appear to an observer on the spacecraft? Justify your answer. (<i>c is velocity of light</i>)			3+2
4.(a) How does radiation interact with matter, and what are the possible types of reactions? (b) Which particles—electrons, protons, or neutrons—participate in nuclear reactions? Justify your answer.			3+2